



Assignment_01

MAT216_Sec_16_23

Summer 2024

Remarks: You will have to submit your assignment on 25 June 2024 (Tuesday, Class time). No late submission will be accepted. Total marks is 40. It will be converted to 20.

1. Use row reduction to transform the matrix in *RREF*. [10]

$$A = \begin{bmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix}$$

2. Mr. X and Mr. Y rent their ice out to the community whenever possible. Mr. X charges [10]
a flat rate of \$20.00 plus \$15.00 for every hour rented. Mr. Y charges \$50.00 for a flat rate
but only asks for \$10.00 for every hour rented.
- (i) Write an equation to model the cost of renting the ice surface for each arena?
 - (ii) Determine the intersection point of the equations. What does this intersection point represent?
 - (iii) Explain when it is best to use Mr. X's service and when it is best to use Mr. Y's service.

3. Determine whether there exists a non-singular matrix A if $A^4 = ABA^2 + 2A^3$, [10]

Where B is the following matrix: $B = \begin{bmatrix} -1 & 1 & -1 \\ 0 & -1 & 0 \\ 2 & 1 & -4 \end{bmatrix}$.

If such a nonsingular matrix A exists, find the inverse matrix A^{-1} .

4. Find the general solution of the homogeneous system $Ax = 0$, where [10]

$$A = \begin{bmatrix} 1 & 2 & 2 & 1 & 4 \\ 3 & 7 & 7 & 3 & 13 \\ 2 & 5 & 5 & 2 & 9 \end{bmatrix}.$$